

Sr DevOps/ DevSecops/ SRE Engineer

Professional Summary:

- Sr DevOps/ DevSecops Engineer with 10+ years of experience spanning DevSecops, Site Reliability Engineering (SRE) and Cloud Engineering. Skilled in designing and managing secure, scalable infrastructure, automating CI/CD pipelines and ensuring high availability across cloud and on-premises environments. Proven ability to enhance system reliability, optimizes performance and collaborates with cross-functional teams to drive innovation and accelerate software delivery.
- Implemented CI/CD pipelines involving Git, Jenkins, Gradle, Maven, Docker, and Ansible for complete automation from commit to deployment. Managed Docker containerization and orchestration using Kubernetes.
- Used Kubernetes to orchestrate the deployment, scaling, and management of Docker Containers.
- Experienced in migrating servers from on-premises to cloud (Azure and AWS). Hands-on experience with databases (MongoDB, MSSQL, Kafka) creating users, performing backup/ restore, and automated snapshots of the VM's.
- Experienced in designing and implementing secure continuous integration and continuous deployment (CI/CD) pipelines, integrating automated security testing (such as static code analysis, dynamic application security testing, and software composition analysis) into the build and deployment process.
- Experience working with Source code management tools like SVN, and GIT and used Automation tool like Ansible for configurations.
- Knowledge of Software Development Life Cycle Methodologies Waterfall, and Agile. Strong work ethic with good Interpersonal and Organizational skills. Ability to work under minimal supervision and guidance.
- Integrated Static application security Testing (SAST) and Dynamic Application Security Testing (DAST) seamlessly into CI/CD pipelines, ensuring automated and continuous security assessments throughout the development lifecycle. Implemented SAST for early identification of code-level vulnerabilities and DAST for runtime security assessment in CI/CD, enabling swift remediation actions and fostering a proactive security culture within development teams.
- Automated deployment activities in middleware environments which includes portal applications running on Java/AWS using Puppet master and Agent nodes and Integrated Puppet with Apache and Passenger.
- Analysed DAST findings and worked with development and operations teams to address identified issues.
- Handled Kubernetes to deploy scale, load balance, scale and manage docker containers with multiple namespaced versions.
- Applied Site Reliability Engineering (SRE) principles to improve system reliability, automate operations and optimize infrastructure costs.
- Generated detailed reports on identified vulnerabilities, including risk ratings and remediation steps.
- Leveraged GitLab for managing Infrastructure as Code (IaC) workflows, using Terraform or AWS CloudFormation, enabling version-controlled and automated infrastructure provisioning.
- Experienced with deployment automation of all the microservices to pull images from the private Docker Registry and deploy to Docker's warm cluster using Ansible.
- Expertise in setting up (CI/CD) Continuous integration and Deployment Pipeline for J2EE, NodeJS and Python applications as well as Docker Containers using tools such as Jenkins, Team city, Bamboo and AWS Code Pipeline.
- Proficient in managing and organizing Git repositories within GitLab, including branching strategies, merge requests, code reviews, and enforcing best practices for version control.
- Extensive experience in building artifacts for Java, Spring framework, Angular applications using ANT, Maven, Gradle, and all required dependencies in Jenkins and AWS Code Build.
- Worked on analysis and visualization of data using ELK (Elasticsearch, Logstash and Kibana) and MySQL databases in a Linux (CentOS) environment.
- Integrated OpenShift with external systems and tools, such as monitoring and alerting systems, to enhance operational efficiency.
- Expertise in Microsoft Azure Cloud Services (PaaS & IaaS), Application Insights, Document DB, Internet of Things (IoT), Azure Monitoring, Key Vault, Visual Studio Online (VSO) and SQL Azure.
- Used JIRA as a Change Management/ Work Management/ SCRUM Agile Tool.
- Expertise in Python, Perl, shell scripting, Groovy to automate administration tasks.
- Experience on Deploying the Software builds to Dev, Stage and Production servers.
- Experts in creating and managing user accounts, security, rights, disk space and process monitoring in Solaris and Red Hat Linux.
- Organized and facilitated project planning, daily stand-up meetings, reviews, retrospectives, sprint, release planning, demos, and other Scrum-related meetings.

Technical Skills:

Operating System : UNIX (HP-UX, Solaris, IBM AIX), Linux (Red hat, CentOS, Ubuntu, SUSE), iOS, Windows Variants.

IAC : Ansible, Terraform, Chef, Puppet

Build Tools : Jenkins, Puppet, Chef, Ansible, Ant, Maven, Perl, HUDSON, BAMBOO, JIRA, AWS, VM Ware, Docker, Vagrant, XL Deploy, XL Release, Enterprise, RedHat, Code Hub, Sonar Qube, Cucumber, SOAP UI, Smoke Test, Rally DEV- CA Agile Central, Sauce labs, Repo, JFrog, Artifactory.

Continuous Integrations : Jenkins, CloudBees Autobahn

Continuous Deployment : Jenkins, Argo CD, GitLab CI/CD

Containerization : Docker and Kubernetes, Rancher

Version Control Tools : GIT, GitHub, Bitbucket, GitLab

Programming Languages : Python, Shell Script, Java

Databases : Oracle, MySQL, PostgreSQL, MongoDB

Ticketing Tool : Service Now, Jira

Web Servers : Tomcat, WebSphere, Web Logic, Jboss, Apache Tomcat, Apache, NGINX.

Testing Technologies : Junit, Selenium, Cucumber, NUnit

AWS Services : EC2, EBS, ECS, ELB, Route53, S3, Lambda, SNS, RDS, EKS, ECR, Cloud Watch, Code Pipeline, Code Deploy, IAM, VPC

Azure Services : Azure Pipeline, Azure Repos, AKS, VM's, Active Directory

Professional Experience:

Samsung SDS America Inc, Ridgefield Park, NJ
Sr. Devops/ DevSecops/ SRE Engineer

Jul 2023– Present

Responsibilities:

- Designed and deployed multi-tier Java and Python RESTful web applications on a 3 tier VPC architecture with high availability distributed across multi availability zones.
- Implemented automatic security testing, including static code analysis (SAST), dynamic application security testing (DAST), Software Composition Analysis (SCA) and container security scanning using tools like SonarQube, Snyk, Black Duck, Veracode DAST, OWASP ZAP, AQUA, Trivy.
- Designed and implemented RESTful APIs using Django and Flask frameworks, enabling seamless integration with frontend applications and third-party services.
- Collaborated with cross-functional teams to design and deploy cutting-edge SRE solutions, aligning with organizational and industry standards.
- Led SRE initiatives, implementing observability tools (Datadog, Prometheus), optimizing SLOs/SLAs, and automating incident responses to improve platform reliability.
- Provided thought leadership on SRE methodologies, promoting continuous improvement and adherence to industry standards.
- Drove transformational initiatives to integrate SRE principles into existing IT operations, fostering a culture of engineering excellence.
- Deployed applications utilizing almost all the AWS stack including EC2, ECS, RDS, S3, DynamoDB, SNS, IAM, Route 53, Load Balancer, etc. using Terraform as infrastructure as code.
- Implemented blue-green and canary deployment strategies with ArgoCD to minimize downtime during releases and improve reliability.
- Worked closely with data scientists to optimize and containerize machine learning models, improving portability and reproducibility across different environments.
- Implemented Terragrunt wrapper scripts to automate common tasks such as environment initialization, state management, and remote backend configuration.
- Created pipelines for clients to subscribe to the Amazon SNS/ SQS topic and receive published messages using a supported endpoint type, such as Amazon Kinesis Data Firehose, Amazon SQS, Lambda, HTTP, email, mobile push notifications, and mobile text messages.
- Implemented infrastructure monitoring and logging solutions using Prometheus, Grafana, and ELK stack to ensure high availability and performance of critical systems.
- Usage of Identity Access Management service (IAM) in creating and managing the user accounts and groups and their policies.
- Implemented and maintained the monitoring and alerting of production and corporate servers/storage using AWS Cloud watch.
- Configure and maintain cluster-level security, such as RBAC (Role-Based Access Control) and network policies.
- Deployed microservice applications on EKS and Kubernetes cluster with persistent volume and ingress controller.
- Managed Kubernetes charts using Helm, and created reproducible builds of the Kubernetes applications, managed Kubernetes deployment and service files and managed releases of Helm packages.
- Worked on Load Balancer security configurations, implementing SSL termination and ensuring adherence to industry best practices.
- Developed a custom Python script for automating routine server maintenance tasks, reducing downtime by 20%.

- Scripted custom solutions in Python for log aggregation and analysis, enhancing real-time monitoring and troubleshooting.
- Trigger CI/CD pipeline jobs automatically using Webhooks, remote triggers and increment application version dynamically in Jenkins and GitHub Action.
- Utilized Open Telemetry to set up centralized logging and metrics pipelines, streamlining log analysis and metric visualization for proactive issue detection and resolution.
- Implementing and managing Splunk for log aggregation, indexing, and analysis to gain insights into system behavior and detect anomalies.
- Designed AWS architecture, Cloud migration, Dynamo DB and event processing using Lambda function, SQS and SNS.
- Maintained Black Duck scan reports and ensured the application code packages were not subject to vulnerability.
- Usage of Launch configurations and Auto Scaling to provide high availability to EC2 machines and effectively used scaling policies based on web traffic inside VPC.
- Implemented Application load balancer (ALB) with Context Path, Host Header and query parameter-based routing.

Environment: Jenkins, Ansible, Bash, Python, GitHub, GitHub Action, Linux, Docker, Kubernetes, Elasticsearch, SonarQube, Splunk, Apache Tomcat, Maven, Gradle, Argo CD, Hadoop, SQL, Tableau, MySQL, Black Duck Scan

Elevance Health, Richmond, VA
SRE Engineer/ Devops Engineer

Jan 2021 – Jun 2023

Responsibilities:

- Developed, maintained, and supported solutions that make it easier for application development teams to compile, deploy, and test their code in AWS.
- Developed and maintained project infrastructure, and managed environments tailored to building, releasing, developing, and testing applications efficiently without interruption to development work.
- Monitored dashboards, analysed logs and metrics to identify potential improvements for our automation systems, and diagnosed build failures.
- Installed, Configured, Managed Monitoring Tools such as Splunk, Nagios and forwarded system stats to Splunk indexer, also centralized monitoring and logging for the systems that are running on cloud(s) and on-premise, using tools such as Nagios and Splunk and Installed monitoring tools Nagios and Splunk.
- Developed, operated, monitored, and automated team infrastructure tools and services, both on-prem and in AWS.
- Actively participated in capacity planning, scale testing and disaster recovery exercises.
- Designed, implemented, and maintained a modern system deployment to the cloud environment with a Continuous Integration/Continuous Deployment (CI/CD) workflow for system build, test and deploy in containerized microservices.
- Developed and Enhanced DevOps operations in multiple Cloud Service environments.
- Built custom tools for employees, empowering them to solve complex problems on their own
- Worked closely with global software development teams and infrastructure teams in Designing and implementing new solutions to streamline manual operations.
- Collaborated with a matrixed team of expert network architects and engineers to drive adoption of DevOps/SRE practices and operating models across all network product owners.
- Developed simple solutions to complex problems, driving the engineering and operations for network software development, automation, provisioning, performance management and tools / APIs for self-service across the enterprise.
- Accountable for establishing technical level relationships and partnering with all business disciplines, outsourcing vendors, and external suppliers to define and implement the Network DevOps / SRE roadmap, building and enforcing the standards defined in it.
- Experienced in managing on-prem (Linux, Windows systems) as well as cloud environment with Ansible Playbooks and Roles. Have built around 50 Ansible Roles to support apps running in environment.
- Collaborate with other Network teams to develop network DevOps/SRE solutions with a focus on production integration
- Conducted network analysis, configuration management and developed improvements for system software performance, availability, and reliability.
- Diagnosed and debugged production incidents and then improved systems to prevent the problem from recurring.
- Was part of a 24/7 On Call support team and collaborate with software development teams.
- Worked on complex problems in a large-scale internal private cloud including observability, monitoring, infrastructure as code, and many other development efforts.
- Ensured that the services meet stability, performance, and availability requirements
- As a Site Reliability Engineering, worked with Engineering teams and provided scalable and robust service-oriented infrastructure automation, monitoring, and analytics solutions for internal private cloud

Environment: AWS, Azure, Apache Tomcat, CloudWatch, CloudFormation, Ansible, Linux, Jenkins, Python, Maven, Gradle, Docker, WebLogic, Nexus, Splunk, Kubernetes, OpenShift, ArgoCD, Open Telemetry

Vanguard, Malvern, PA
DevOps/Cloud Engineer

Sep 2017 – Dec 2020

Responsibilities:

- Worked on AWS, Azure and Terraform API to create physical resources on the cloud utilizing infrastructure as a code, spinning up and destroying resources as necessary.
- Created AWS resources like VPC, subnets, EC2 instances, ELB, security groups, S3 by using Terraform.
- Automated Weekly releases with Maven scripting and Compiling Java code, Debugging and Placing Builds into Maven repository. Responsible for designing and deploying best SCM processes and procedures.
- Creating and developing deployment, namespace, Pod, Service, ConfigMap, StatefulSet, Health Checks etc. for Kubernetes in YAML Language.
- Set up and manage TLS/SSL certificates for securing traffic, including certificate issuance, renewal, and rotation.
- Utilized Kubernetes for the runtime environment of the CI/CD system to build, test deploy.
- Implemented a production ready, load balanced, highly available, fault tolerant Kubernetes infrastructure.
- Deploying the WebSphere Liberty operator application Kubernetes and OpenShift platforms.
- Implementation of IBM WebSphere Liberty for Amazon EKS on AWS environment.
- Implementation and Administration of ELK stack for Application Logs centralization and Grafana for Monitoring.
- Responsible for creating Terraform scripts for deploying various resources in AWS for various environments.
- Involved in Blue/green deployment strategy by creating identical applications, which already existed in the production environment.
- Performs 24/7 production/test support to perform Break/Fix Analysis/Troubleshooting in servers.
- Involved in AWS EC2/VPC/S3/SQS/SNS based automation through Terraform, Ansible, Python and CloudFormation. Adopted new features as they were released by Amazon.
- Working on AWS provisioning AWS EC2 Infrastructure and deploying applications on AWS Beanstalk (EBS), ECS Fargate and EKS.
- Built Jenkins jobs to create AWS infrastructure from GitHub repos containing Terraform code and administered/engineered Jenkins for managing weekly builds.
- Used GIT version control to manage the source code and integrating GIT with Jenkins to support build automation and integrated with JIRA to monitor the commits.
- Configured and managed source code using GIT and resolved code merging conflicts in collaboration with application developers.
- Integrated Elasticsearch with other components of the ELK (Elasticsearch, Logstash, Kibana) stack to create comprehensive log management and monitoring solutions.
- Coordinating Release effort amongst various teams (Development, QA, Testing, and Business Analysis) in geographically separated environment and supported and developed tools for integration, automated testing and release management.
- Saved remarkable time and cost of testing by automation using Selenium WebDriver, RUBY, Cucumber
- Used Jenkins and Nexus to build code from Subversion and deploy to testing and developing environments.

Environment: AWS, EC2, RDS, Python, DynamoDB, S3 bucket, Route53, Docker, GIT, Jenkins, Kubernetes, Ansible, Terraform, Bash Script, WebSphere Liberty, Elasticsearch, Maven, Gradle, Nginx

OSI Digital, Hyderabad, India
Application Release Engineer

Aug 2014 – Sep 2016

Responsibilities:

- Maintained and enhanced existing PUPPET modules to be deployed across various providers and deployment architectures.
- Managed PUPPET classes, resources, packages, nodes and other common tasks using PUPPET console dashboard and live management.
- Creating & conducting walkthroughs of 'Deployment Plans' for complex large-scale Releases including Mock Deployments in Pre-Prod environments & Production Installations.
- Working with an agile development team to deliver an end-to-end continuous integration/ delivery product using Run deck integrated with JENKINS to get the job done.
- Deployed PUPPET Dashboard and PUPPET DB for configuration management to existing infrastructure.
- Involved in migration activities of Java scripts and Database scripts from Oracle, MS SQL Server and MYSQL into different environments like Development, QA, UAT and Production on Red Hat Enterprise Linux (RHEL) Infrastructure.
- Designed and implemented GIT metadata including elements, labels, attributes, triggers, and hyperlinks.
- Implemented & maintained the branching and build/release strategies utilizing GIT.
- Performed all necessary day-to-day GIT support for different projects.
- Responsible for design and maintenance of the GIT Repositories, and the access control strategies.

- Experienced with CI tools (JENKINS) and Version Control Tools or Source Code tools.
- Responsible for writing Manifests to configure nodes.
- Experienced in building automation using JENKINS, Maven, Gradle, ANT.
- Experienced in deploying Java projects using Maven/ANT and JENKINS.
- Good understanding of building Android Applications using Maven, Gradle and JENKINS.
- Building post install scripts using Shell scripting in Linux servers.

Environment: Java, J2EE, ANT, Maven, Gradle, JENKINS, Tomcat, GIT, GitHub, Bash, PUPPET, Linux, CentOS, JUnit, Nexus, Oracle, MS SQL Server.

Education:

Master's Degree in Data Science, University of Buffalo – 2017

Bachelors in Technology - VNR Vignana Jyothi Institute of Engineering and Technology – 2014